

Rearranging the terms and removing repeated terms, we get:

$$(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D').$$

$$(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D')$$

$$\Pi(0,2,4,6,7,8,10,11,12,15)$$

Simplified POS:

$$F = (C'D).(C'D').$$

$$(A+B+C).(A+B+C')$$

AB \ CD	00	01	11	10
00	0		0	
01	0		0	0
11	0		0	
10	0		0	0

$$F' = (C'D').(A+B+C'D).$$

$$(A+B+C'D)$$

$$F = F' = (C'D').(A+B+C'D)$$

$$(A+B+C'D)'$$

AB \ CD	00	01	11	10
00		0		0
01		0		
11		0		0
10		0		

$$= (C'D') + (A+B+C'D)' + (A+B+C'D)' \text{ (Extended Theorem 5(b))}$$

$$= (C'D') + (A'.B'.C'.D') + (A').(B').(C').(D')$$

$$\text{(Extended Theorem 5(a))}$$

$$= (C'D) + (A'.B'.C'.D') + (A.B.C.D) \text{ (Theorem 3)}$$

$$= (A+B+D).(A+B+C).(A+B+D')(B+C+D) \text{ (Postulate 2(a))}$$

$$= (A+B+D+0).(A+B+C+0).(A+B+D+0).(B+C+D'+0)$$

$$= (A+B+D+C.C).(A+B+C+D.D).(A+B+D+C.C).(B+C+D'+A.A)$$

$$= (A+B+D+C.C).(A+B+C+D.D).(A+B+D+C.C).(B+C+D'+A.A) \text{ (Postulate 5(b))}$$

$$= (A+B+D+C).(A+B+D'+C).(A+B+C+D).(A+B+C+D) \text{ (Postulate 4(b))}$$

$$(A+B+D+C).(A+B+D'+C).(A+B+C+D).(A+B+C+D)$$

Rearranging the terms and removing repeated terms, we get:

$$(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D).$$

$$(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D).(A+B+C'D)$$

$$\Pi(3,7,6,9,11,13,15)$$